

19	C	Option A: Depending on the charge of the object, the force could either be left (positive charge) or right (negative charge). Option B: The magnitude of the field strength can be found using $E = -dV/dr$. The gradient at P is larger than Q, so the field strength is larger at P Option C: The potential at P is lower than R, and thus ΔV is a negative number. The work done is $W = q \Delta V$ and this is a positive number since q is also negative. So, the work done is positive $\square \square$ Option D: The potential energy is given by $U = qV$, and so, since V is more negative at P than at R, the potential energy is lower at P than at R.
20	D	Using $E = ma$, thus $eE = ma$